

AI & In-Line Suggestions to Improve Student's writing

Design Track - INST 639T

Problem Space

With the introduction of AI in the creative process, over-reliance on AI to make creative outputs such as writing and drawing is a problem slowly creeping in. With Large Language Models (LLMs) getting more advanced, new ways of integrating AI into writing are emerging. With the increasing availability of this capability on various platforms, the risk becomes of AI taking over human thinking, and contributing more than required to a person's personal voice and thought process. The question of agency and creative contribution of AI to a written piece of work is becoming more relevant. Tools such as ChatGPT, Gemini, Grammarly etc are common 'assistants' that writers go to for ideation, editing, co-writing, and cleaning up their writing. The default nature of these tools is to provide a 'finished output', that is ready to use. A user can paste their written body of text, and the AI generates a polished version of it in a few seconds. With some focused prompts, this text can be turned into any style of writing, number of words, tone, or perspective. Most of these tools are 'message' based, or designed in a 'Chat' based format, where the anthropomorphic quality of 'talking' to an AI is used as the primary interaction pattern. However, there are other ways that AI can be designed, for example - Auto-filling text (Gmail, Google Docs) or highlighting text and offering suggestions as one is writing (Autocomplete on iPhone, AutoCorrect, Grammarly). The activity of writing can also be studied as broken down into many cognitive phases such as - Ideation, Editing, Translation, Monitoring, Reviewing etc. as mentioned in various frameworks considered in Reza et. al (2025). Support from AI, depending on which of these activities is targeted, can influence the way writers perceive their agency and creative confidence in writing.

Specific User & Context

Another layer of interest comes in when a user, or a specific context is considered. Writing Academic Articles, Emails, Text Messages, Poetry, Blogs, Assignments - all of

these different contexts have different needs. An interesting space that makes AI impactful in the process of writing is for Students - Since writing is the basic medium that educators use to assess whether a student has adequately absorbed the class material, the invention of readily available AI tools have forced educational institutions to make policies around use of AI in daily homework assignments, and made teachers change the way they assign work, so that AI cannot be used to cheat, or do the work on the student's behalf. This Project picks the common creative task - writing - as the medium, and proposes an AI tool that helps students in the 'process' of writing, rather than providing end outputs that can be copy-pasted.

Design Rationale

AI as a collaborator can have positive effects on student's writing - coming up with many unique ideas and improving quality of writing as opposed to writing just manually, without any AI input as shown in Doshi & Hauser (2024). With this technology becoming more available, it is undeniable that students will turn to help from AI in times of need. Tools such as NotebookLM are designed specifically for students to understand complex material such as text books and research articles. However, writing with AI can be strictly prohibited for assignments, as the work of thinking, articulating and putting thoughts on to paper is the work that helps students learn the subject material. It is a valuable space to look into designing AI tools that don't do the student's thinking work for them, but instead support their learning.

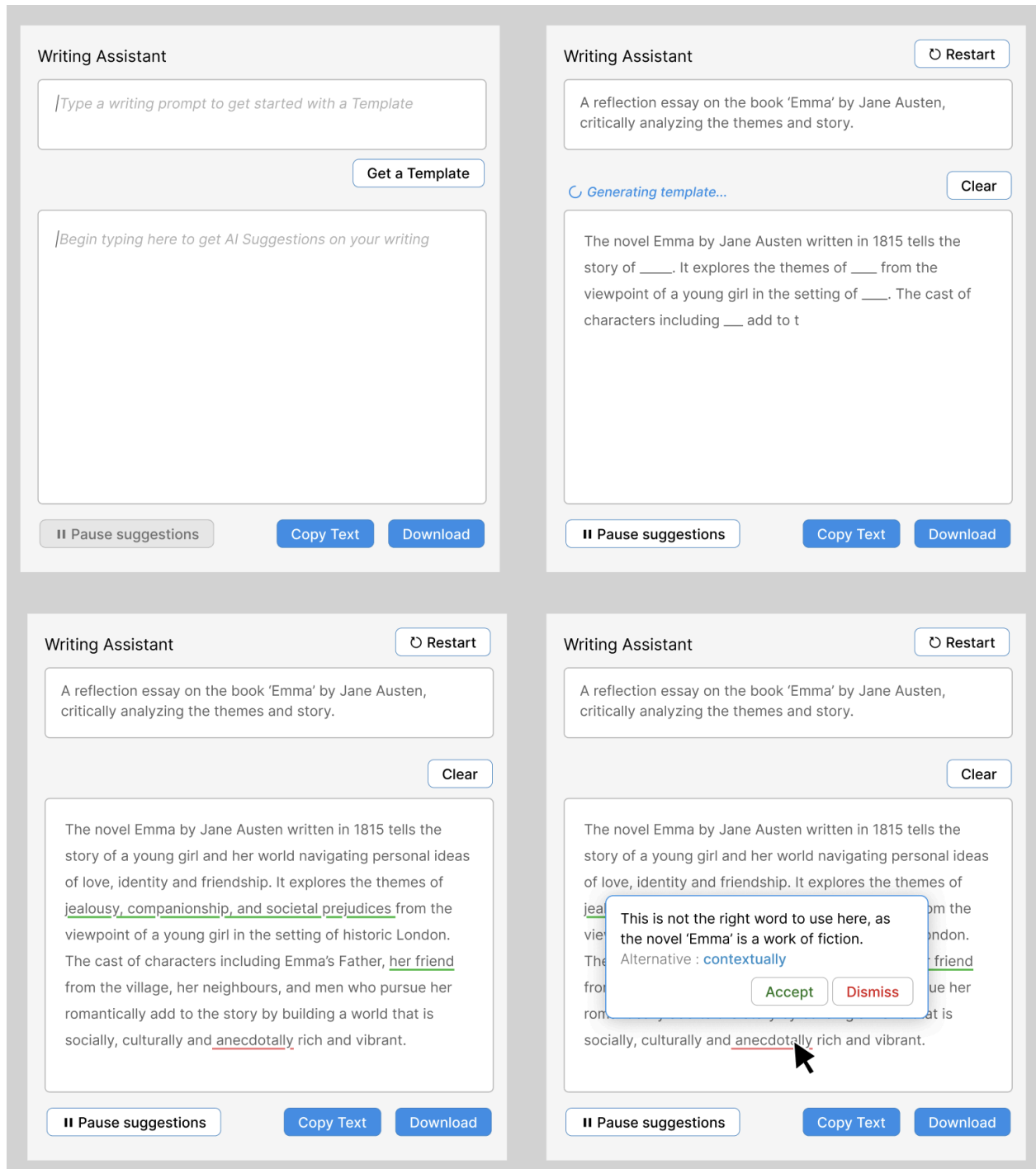
Through interface design, the positive effects of using AI tools can be amplified, while design strategies can be used to reduce effects that hamper creativity and increase over-reliance on AI. When it comes to creative tasks, humans viewing AI tools as a lower social referent (Reich & Teeny (2026)) puts AI in a unique position to help improve creative confidence and provide a push, a launching pad, or get rid of the fear of a 'blank canvas' or a writer's block, without bringing down the morale of the writer, especially in more complicated writing tasks.

Prior research on the subject of message level & sentence-level feedback on writing shows that Message level (Chat-based interfaces) improve efficiency in writing, whereas sentence level suggestions improve original content creation Fu et. al (2023). Sentence level or In-line suggestions make the process of writing slower, as the writer is thinking about every suggestion from the AI and making a decision to accept the change or not,

but in the process engaging deeply with the piece of writing. Fu et al. (2023) mentions some technical considerations that can help in the design of such tools. The 'Unit of Suggestion' tells at which sentence lengths suggestions are provided - few words, phrases, multiple sentences or paragraphs. Suggestions of re-writing larger paragraphs show reduced perception of agency, and shift of writers into editing mode, rather than drafting / writing original content. Another consideration was showing different options in one suggestion for a writer to choose from - this can be tricky to manage because of limited visual space of sentence-level suggestions, and the risk of longer suggestions becoming distracting.

Proposed Solution

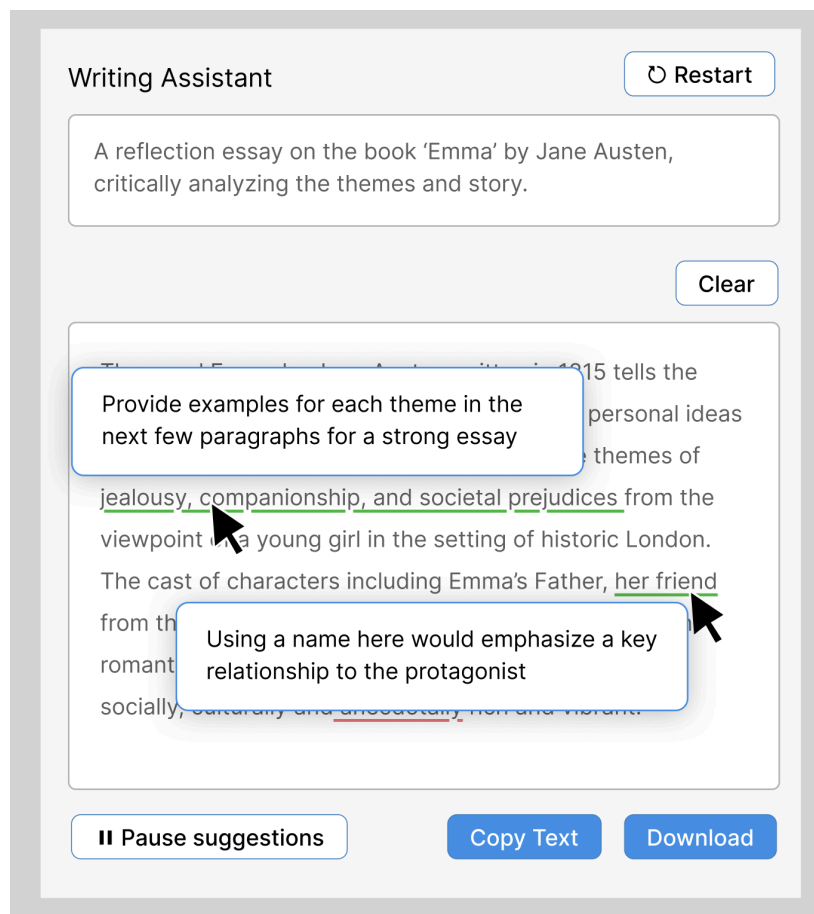
The proposed solution here is an interface that aids writing - in the process of critical thinking. Instead of providing an end output, it highlights areas that can be improved, shows edits that can be made to improve it, and provides ideas as one writes. This format is similar to how miss-spelled or wrong words are underlined as one types in a Google Doc or Microsoft Word document. The AI can be turned on or off - giving students the flexibility to switch from manual to AI mode as they wish. The first output from the prompt is a template in an editable form. Settings to adjust Tone, Pacing, bringing in references from online sources etc. can enhance the experience further. Taking insights from frameworks mentioned in Reza [et.al](#) (2025), the design prioritizes Human agency in different parts of the writing process like planning and reviewing. When compared to tools already present in the market, this interface is designed to assist students, not take over the writing for them. Minimal 'editing' from the AI in changing words or phrases promotes the voice of the student to be prioritized. To get rid of the fear of a 'blank canvas', starter templates depending on the brief are provided to help students get started with writing. The AI Suggestions nudge students to think about 'what to write' - different perspectives they could include, where to add examples, any points they are missing, the structure of their essay, and the logical flow of their arguments. This goes beyond checking grammar and checking spelling mistakes of traditional writing tools, but does not hamper the thinking process of a student, or provide any new ideas or materials directly to include in the writing.



(Above) Screens showing different states of interaction, and AI Suggestions. The process of writing a prompt, filling out a template and receiving suggestions.

For example, for a question “Write a reflection essay on the book ‘Emma’ by Jane Austen, critically analyzing the themes and story.” The first output to this would be a

template such as “ The novel Emma by Jane Austen written in 1815 tells the story of _____. It explores the themes of _____ from the viewpoint of a young girl in the setting of _____. The cast of characters including _____ add to the story by _____”. Here, there is no addition of viewpoint or opinion, but just a way to get a student started on ‘what to write’. It is suggested that the storyline, the setting and different characters can be spoken about to point out themes in the novel. This is generally the task of a Teacher, Professor or Teaching Assistant providing guidance to students in study halls or Office Hours. Additional prompts in the duration of writing would be along the lines of “Are there any other sub-themes to mention?” , “Provide an example to support this opinion here?” , “ This point has been mentioned 3 times”, “The structure of this paragraph is changing contexts two times, simplify it”, “This paragraph can be broken into two topics with separate headings”. The prompts offer support in the skill of writing as a reflection of thinking critically about one’s exploration of a subject.



(Above) Suggestions of different types on the given example text.

Evaluation Plan

The main aim of evaluating this design is to compare the effectiveness of this tool to a general chat-based AI tool such as ChatGPT or Gemini. Participants would be asked to rate their satisfaction with their end written output, their confidence in the tool, their confidence in their output, and their own writing skills. The main method of study would be to recruit students from High School and Early College years of Age group 15-20 years old, and provide neutral topics of writing. Topics can be chosen from a subject students are familiar with such as English, Biology or History etc. Students are divided into two groups, and provided the tools. Group one receives the Sentence Level Tool of this Project , and the other group receives a Chat based tool such as ChatGPT to write with. The students are given a time of 45-60 minutes to write a few paragraphs (500 - 700 words) on the topic. They then fill out a questionnaire of their experience using the tool, and answering the previously mentioned questions on a Likert Scale, and free form text answers. As a follow up, outputs from these participants would be 'graded' by evaluators. Rating of the quality of writing from Evaluators would provide insight into the objective quality of the writing from using this tool, and comparing it to chat-based interfaces. The evaluators consist of a mix of teachers and professors, they are asked to grade these essays, as they would for their own classes. The recruitment of participants for this study can be possible through a snow-ball method of reaching out to students, collaboration with an educational Institute, or through platforms such as Prolific.

Ethical & Equity Considerations

Two questions that take center stage for a tool like this are - Will this tool be accepted by teachers? And will this tool have inherent biases that push students into a specific mindset or style of writing? Since the dawn of AI writing products in the market, it is not feasible to monitor students for all submitted assignments or writing activities for AI use. With the easy availability and proven positive effects of these tools, students will turn to them to understand, write and make sense of study material. However, at what point, or at which AI features will the line be drawn in the educational space? This tool has the potential to be adapted by schools and colleges as a standard in taking AI help, and the limit to which AI can be used for writing. Companies that create educational tools like ELMS, Turnitin and AI checkers inbuilt into administrative applications used in Schools and Colleges can benefit from using an application like this to provide an integrated, yet monitored assignment writing experience for students. For original intellectual material such as research papers or articles, authorship becomes a greater concern. How will the

rules of AI use differ for students, and should credit be given to such a tool that helps in critical reflection of academic material, and prompts more than basic English grammar and structure? Another concern comes into light when different subjects are considered. Politically charged topics in subjects such as History, Social Sciences, Communications, Health and Psychology - will AI have inherent biases because of training data? Will it show in the prompts given to students, or hint subtly at certain sides of opinions? For subjective opinions, and subjects where a perspective on a topic becomes critical to forming the student's viewpoint, writing nudges may influence students. A follow up study for special topics would help uncover this - where different Code bases, or APIs are used to check biases in the prompts, and how they affect student's writing.

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